

Paper 3

AI-Assisted Portfolio Triage for Canada's Major Projects: A Transparent, Auditable Forecasting Approach

Running Title: AI Portfolio Triage – Forecasting Approach

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Abstract

Canada's Major Projects Inventory (MPI) provides seven year-end snapshots of 1,250 unique projects and more than 4,000 project-year observations from 2017 to 2024, but offers little forward visibility. We introduce a lightweight, auditable forecasting toolkit that pairs a Bayesian likelihood-ratio scorecard, which estimates the probability that a project will advance to construction, with a Cox proportional hazards model, which estimates the time to that outcome. The scorecard begins from a historical prior of 28 percent of new projects reaching construction within five years, and then updates those odds using likelihood ratios drawn from geography, sector, cost, cleantech status, and other descriptors. Location and sector emerge as the strongest predictors, while cost and cleantech status add only modest lift. The scorecard achieved an AUC of 0.81 and the Cox model a concordance index of 0.79, meeting accepted benchmarks for functional accuracy. Together they generate a unified risk surface that reveals both probability and timeline for every active project, enabling market participants and planners to triage portfolios, stress-test policy, and allocate resources with greater confidence. The framework's transparency also supports routine recalibration and extension across Canada's diverse sectors.

Introduction

Paper 1 and Paper 2 use descriptive statistics, cohort tracking, and transition analysis to evaluate backward-looking, project-level advancement and outcomes using Canada's Major Projects Inventory (MPI). This paper poses two straightforward questions of the MPI dataset: "Will it build?" and "How much longer?" The main technical hurdle is reconciling the dataset's diversity while keeping the model lightweight, transparent, usable by non-specialists, and functionally accurate. We address this challenge by pairing a Bayesian scorecard, which turns categorical clues into an intuitive build probability, with a Cox proportional hazards model that converts the same variables into expected years remaining in development.

The contribution is twofold. First, we deliver Canada's first fully transparent diligence and planning tool that estimates both likelihood and timing of a MPI project advancing to construction, enabling market participants and strategic planners to screen and compare hundreds of records quickly. Second, we show that off-the-shelf statistical methods, reinforced by readily available AI and machine-learning utilities, can meet the threshold for functional triage in finance and medicine. The models make project surveillance and pipeline forecasting open and verifiable, giving decision-makers a common evidence base of clear probabilities and timelines that anyone can audit and replicate.

Methods

Dataset and training

The models were trained and tested on a subset of the MPI. The training cohort comprised 268 projects that first entered the MPI between 2018 and 2020 and were tracked through the 2024

snapshot, by which time 28 percent had advanced to construction. Projects entering after 2020 were treated as right-censored due to insufficient follow-up time.

Each annual MPI record includes a project identifier, sponsor, province or territory, location, sector and group classification, estimated cost, ownership type, a cleantech flag, and a status code such as Announced & Planning, In Review, Approved, or Under Construction. These descriptors were converted into model covariates. For validation, the dataset was partitioned using an 80–20 stratified split that preserved the province–sector distribution across training and test samples.

Model objectives and accuracy thresholds

The study aimed to construct lightweight, auditable risk engines capable of estimating both the probability that a project would break ground and the expected time to that outcome. The design target was “functional” accuracy consistent with standards in finance and medicine.

For binary classification, an area under the receiver operating characteristic curve (AUC) between 0.70 and 0.85 is generally considered functional, with values above 0.85 regarded as strong. Scores below 0.70 require refinement or the use of segment-level overlays, while values near 0.50 indicate performance equivalent to random guessing. For survival analysis, the concordance index (C-index) provides the equivalent measure, with 0.70 widely accepted as the minimum threshold for practical use. The models were designed to achieve mid-0.70s performance, balancing interpretability and predictive strength.

Table 1 summarises the benchmarking conventions used in this study.

Discrimination band	Street label	Typical application	Metric cited
< 0.70	Weak	Emergency flu-triage screen; mass-market marketing response	AUC / C-index
$0.70 - 0.85$	Functional	30-day hospital-readmission risk score; consumer-credit default scorecard	AUC / C-index
> 0.85	Strong	Breast-cancer mammography classifier; real-time payment-card fraud detection	AUC / C-index

Table 1: Model Accuracy Benchmarking Conventions.

Rationale for model selection

A Bayesian scorecard and a Cox proportional hazards model were selected for their transparency, interpretability, and computational efficiency. Gradient-boosted trees and random forests achieved similar AUC values but required extensive hyperparameter tuning and obscured predictor effects. Logistic regression struggled to meet accuracy thresholds unless rare categories were collapsed, which eliminated the regional and sectoral nuance critical to the MPI. The Bayesian likelihood-ratio framework preserved all categories, provided explicit and auditable weightings, and delivered functional accuracy.

Alternative time-to-event methods, including survival forests and neural networks, either reduced concordance accuracy, imposed heavy computational demands, or limited interpretability. The Cox model accommodated right-censoring, produced hazard ratios with direct substantive meaning, and handled categorical covariates with flexibility. The proportional-hazards assumption was tested and found reasonable for the dataset, making the Cox approach suitable for this application.

Bayesian scorecard

The Bayesian scorecard addressed the question “Will it build?”. Projects were classified as successful once they reached Under Construction and unsuccessful if cancelled or removed from the MPI. The model began with a prior probability derived from the 2018–2020 cohort. Each categorical covariate—province, sector, province–sector interaction, sponsor type, cost quintile, cleantech flag, and start year—was converted into a likelihood ratio. Ratios greater than one increased the odds of success, while ratios below one decreased them. Sparse categories were stabilised using Laplace smoothing, and low-count province–sector cells defaulted to neutrality. Likelihood ratios were multiplied and summed on the log-odds scale, allowing contributions to be expressed in bits, where a plus-one bit increment doubled the odds of success and a minus-one bit halved them. The final log-odds were converted to a probability between zero and one. Features with low information value were excluded, and the remaining coefficients were regularised through logistic scaling into a stable point system, ensuring interpretability and auditability.

Cox proportional hazards model

The Cox proportional hazards model addressed the question “How much longer?”. Each project was followed from the year of entry in the MPI until construction, cancellation, or right-censoring at the 2024 snapshot. The covariates mirrored those of the Bayesian scorecard. Project cost was treated as a quintile derived from reported percentiles. To mitigate overfitting, a ridge penalty of 0.10 was applied, reducing coefficient magnitudes without altering their rank order.

Hazard ratios provided interpretable measures of relative event speed. For example, a hazard ratio of 1.33 for natural gas power projects indicated a one-third faster time to construction or cancellation relative to the reference group. Ratios below 1.0 signalled slower progression

through development. Schoenfeld residual tests confirmed that the proportional-hazards assumption held globally, with only a few low-count categories approaching significance.

Model integration and automation

The two models drew on the same covariates and were therefore complementary. The Bayesian scorecard produced a probability of construction, while the Cox model generated a survival function and expected time-to-event. Together, they produced a unified risk surface showing both likelihood and timing.

Model preparation and validation were carried out in Python notebooks. Automation tools assisted with feature binning, hyperparameter sweeps, and diagnostic visualisation. Optuna was used for limited hyperparameter tuning, and SHAP plots confirmed that the same risk drivers were emphasised in both models. These tools supported reproducibility while maintaining transparency in the statistical framework.

Results & Discussion

Bayesian scorecard results

The Bayesian scorecard began with a historical prior: 28 percent of projects first listed in 2018–2020 advanced to Under Construction by 2024, corresponding to baseline odds of about 0.385. Each categorical covariate was converted into a likelihood ratio that indicated its positive or negative influence on success. Laplace smoothing was applied to stabilise sparse categories, and province–sector cells with fewer than two cases were defaulted to neutrality, preserving transparency while avoiding extreme swings.

Figure 1 and Figure 2 present the likelihood-ratio spectrum, which highlights the primacy of geography. Provincial effects, and especially province–sector interactions, produced ratios well above the neutral value of one, showing that location and industry together exert strong influence on project advancement. By comparison, the cleantech flag clustered close to neutrality, offering only limited discriminatory power. Capital-cost quintiles showed a mixed pattern: higher-cost projects were associated with moderately higher odds of success, while lower-cost projects remained close to neutral. Broad sector, sub-sector grouping, and project-start vintage contributed incremental information but rarely rivalled the strength of geographic signals. Overall, where and what is built are more decisive than when or at what scale.

Likelihood Ratios (2-column layout, red dashed = LR 1)

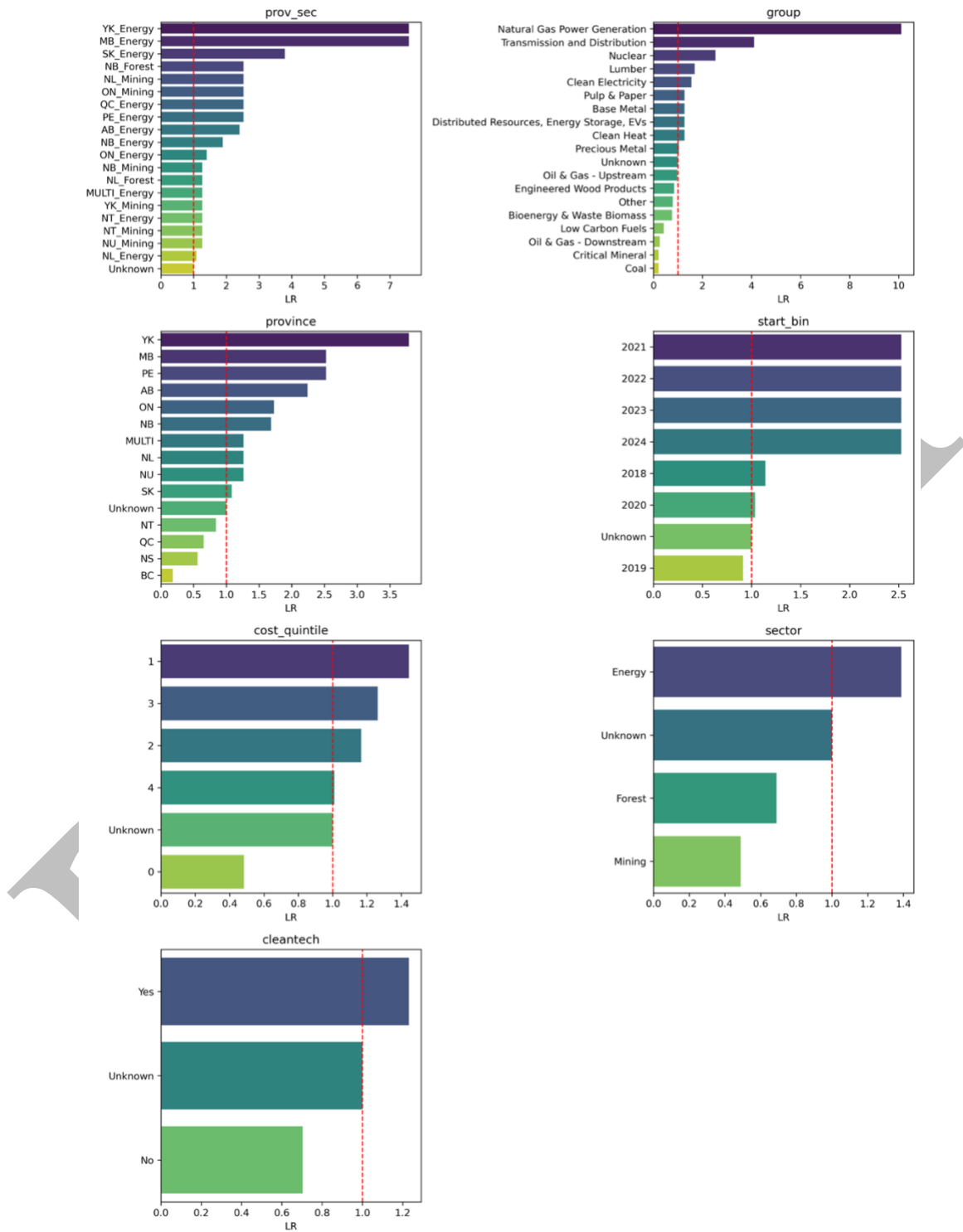


Figure 1: Bayesian Scorecard Likelihood Ratios.

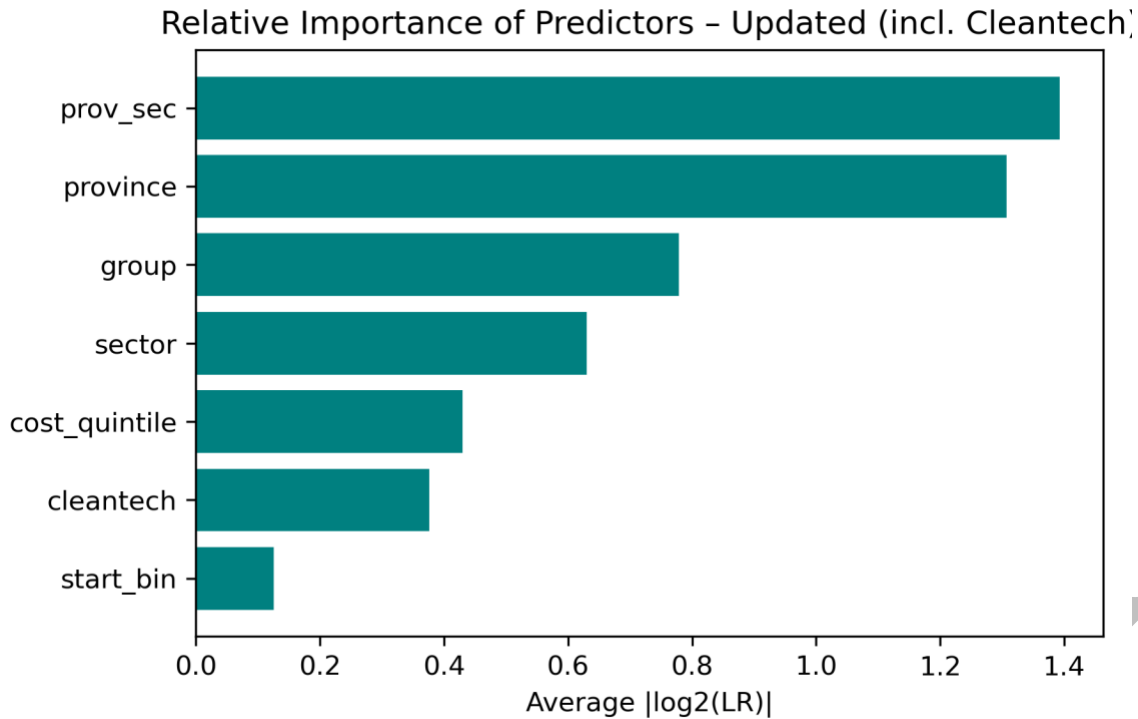


Figure 2: Bayesian Scorecard Relative Importance of Predictors.

Model accuracy was assessed on an 80 percent training sample and a 20 percent hold-out that preserved the province–sector mix. Figure 3 displays the receiver operating characteristic curve for the hold-out sample, with an AUC of 0.81. This value lies well within the functional range defined in Table 1, indicating that the scorecard consistently ranked projects that advanced to construction ahead of those that did not. The steep initial rise of the ROC curve shows that, by accepting roughly 20 percent false positives, the model captured about 70 percent of true positives, a recall rate suitable for triage applications.

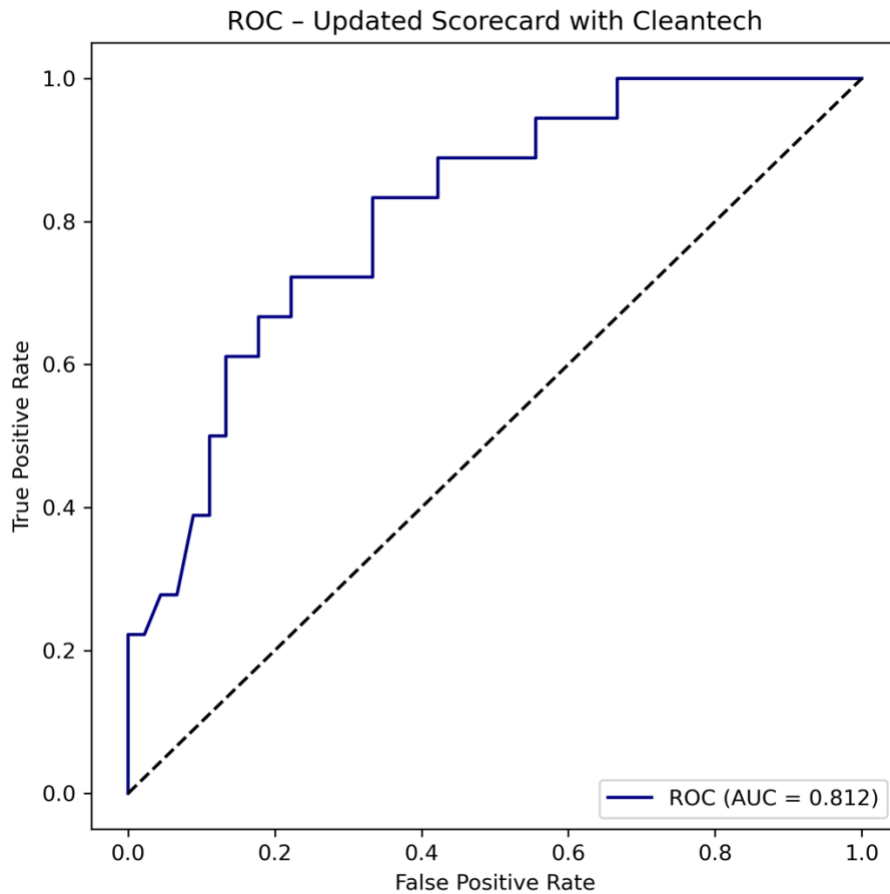


Figure 3: Receiver Operating Characteristic Curve for the Hold-Out Sample.

Because the scorecard aggregates evidence on the log-odds scale, contributions can be expressed in “bits,” where a plus-one bit increment doubles the odds of success and a minus-one bit halves them. This structure enabled project-level interpretation. Figure 4 illustrates two worked examples from the test data. An Alberta coal mine, despite benefiting from a positive provincial factor, faced strong negative signals from the mining sector overall, the coal sub-sector, and the province–sector interaction. These effects, compounded by the absence of a cleantech designation, reduced its predicted probability of success to about 6 percent, or roughly one in 17. By contrast, an Ontario precious metals mine combined a favourable provincial setting with a positive province–sector interaction and higher-cost category, which more than offset sector-

level headwinds. Its predicted probability of success was about 56 percent, slightly better than even odds.

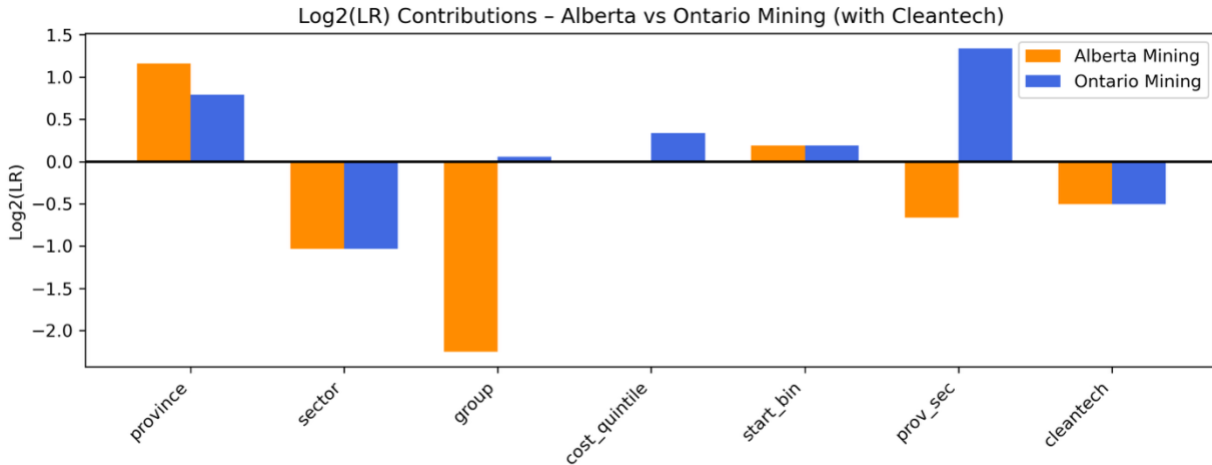


Figure 4: Log-Odds Scale for Two Worked Samples from Dataset.

Cox proportional hazards results

The timing analysis used a Cox proportional hazards model to follow each project from its initial listing in the MPI until it either reached *Under Construction*, was cancelled, or remained right-censored at the 2024 snapshot. The covariates mirrored those of the Bayesian scorecard: province, sector, sponsor type, cost quintile, and cleantech flag, with cost treated as a quintile derived from reported percentiles. A ridge penalty of 0.10 was applied to limit overfitting and multicollinearity, shrinking the largest coefficients modestly without altering their rank order.

When tested on a stratified 20 percent hold-out, the Cox model achieved a concordance index of 0.79, indicating that four out of five project pairs were correctly ordered by completion speed. This performance meets the “functional” benchmark defined in Table 1.

Figure 5 shows the Kaplan–Meier survival curve for the reference case, an Alberta–Energy project with no covariates applied. Attrition began quickly: one quarter of projects exited within three years, half within five years, and three quarters within seven years. This reference hazard function provides a benchmark against which covariate-adjusted trajectories can be interpreted.

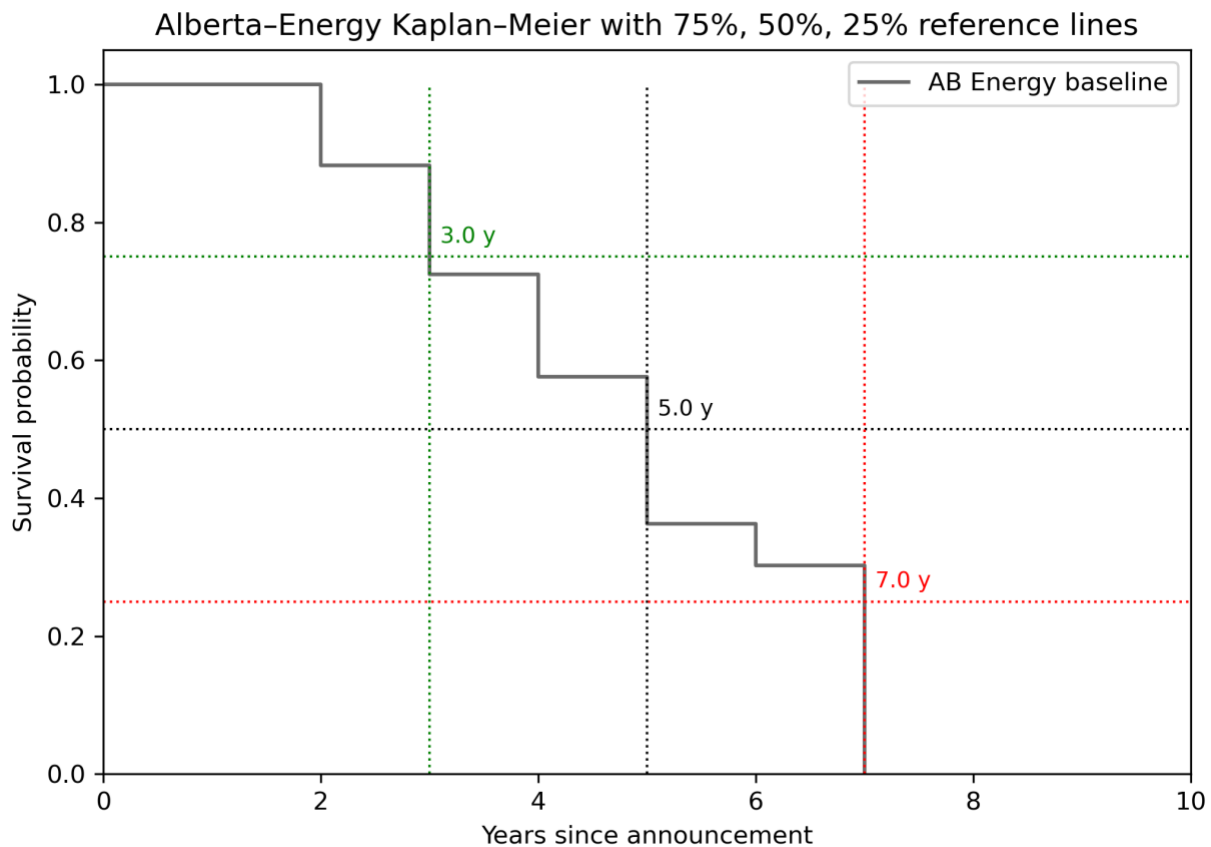


Figure 5: Kaplan–Meier Survival Curve for the Reference Case.

Because the dataset ends at seven years, no Alberta–Energy projects are observed beyond that point and the Kaplan–Meier curve falls to zero. Figure 6 overlays a Weibull fit that smooths this artefact. The dashed orange line marks the “one survivor left” threshold ($1 \div 78$, or approximately 0.013). The dotted red guides indicate where the extrapolated curve intersects this

threshold, suggesting that, if current hazards persist, the final Alberta–Energy project would resolve about 8.5 years after announcement.

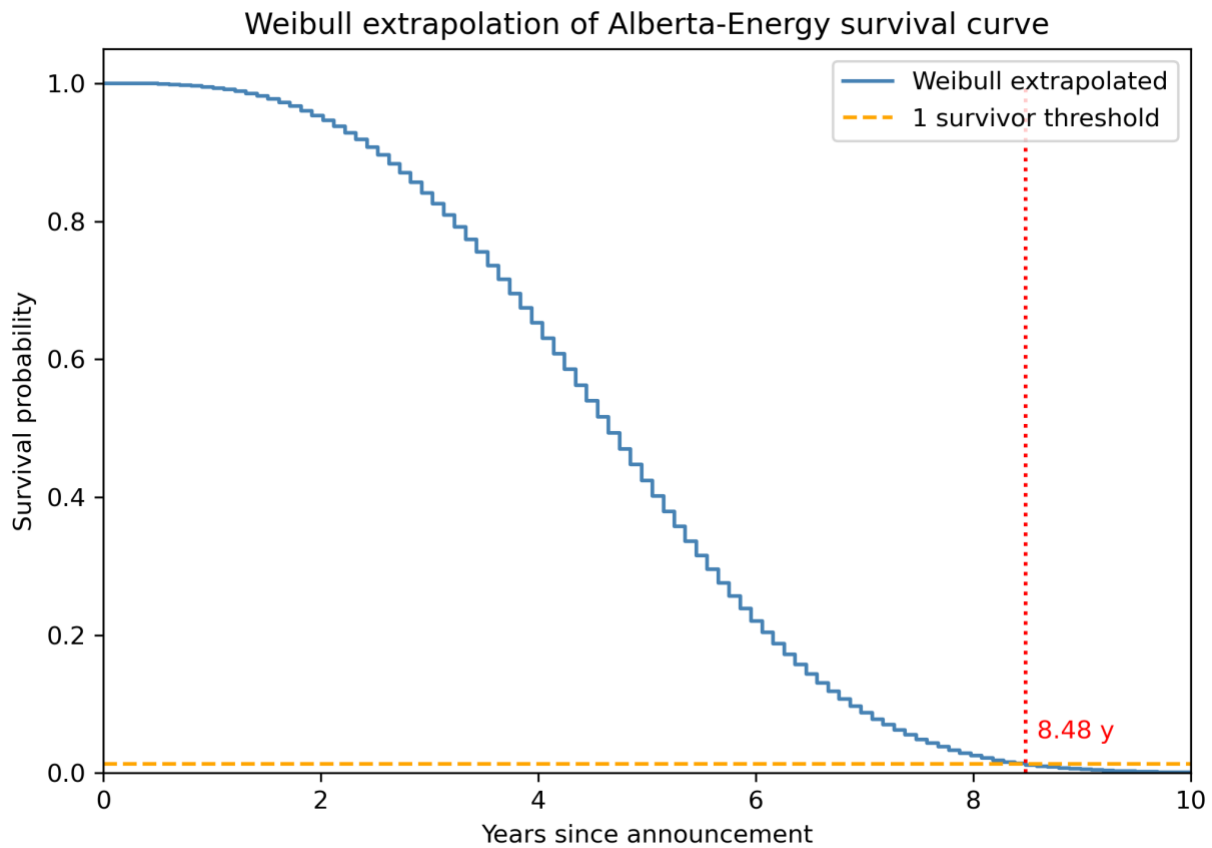


Figure 6: Weibull Curve for the Reference Case.

Figure 7 displays the regularised coefficients for the Cox model, with Alberta–Energy fixed as the reference category. Several provincial effects stand out: British Columbia (hazard ratio ≈ 0.29) and multi-jurisdiction projects (≈ 0.33) survived far longer than the baseline, while Ontario showed a positive coefficient consistent with faster exits. Sectoral effects were similarly differentiated. Mining (hazard ratio ≈ 0.42) exited more slowly, while forestry (≈ 1.34) exited about one-third faster. Cost was effectively neutral, while the cleantech flag raised the hazard ratio to about 1.26, suggesting earlier resolution for green-labelled projects.

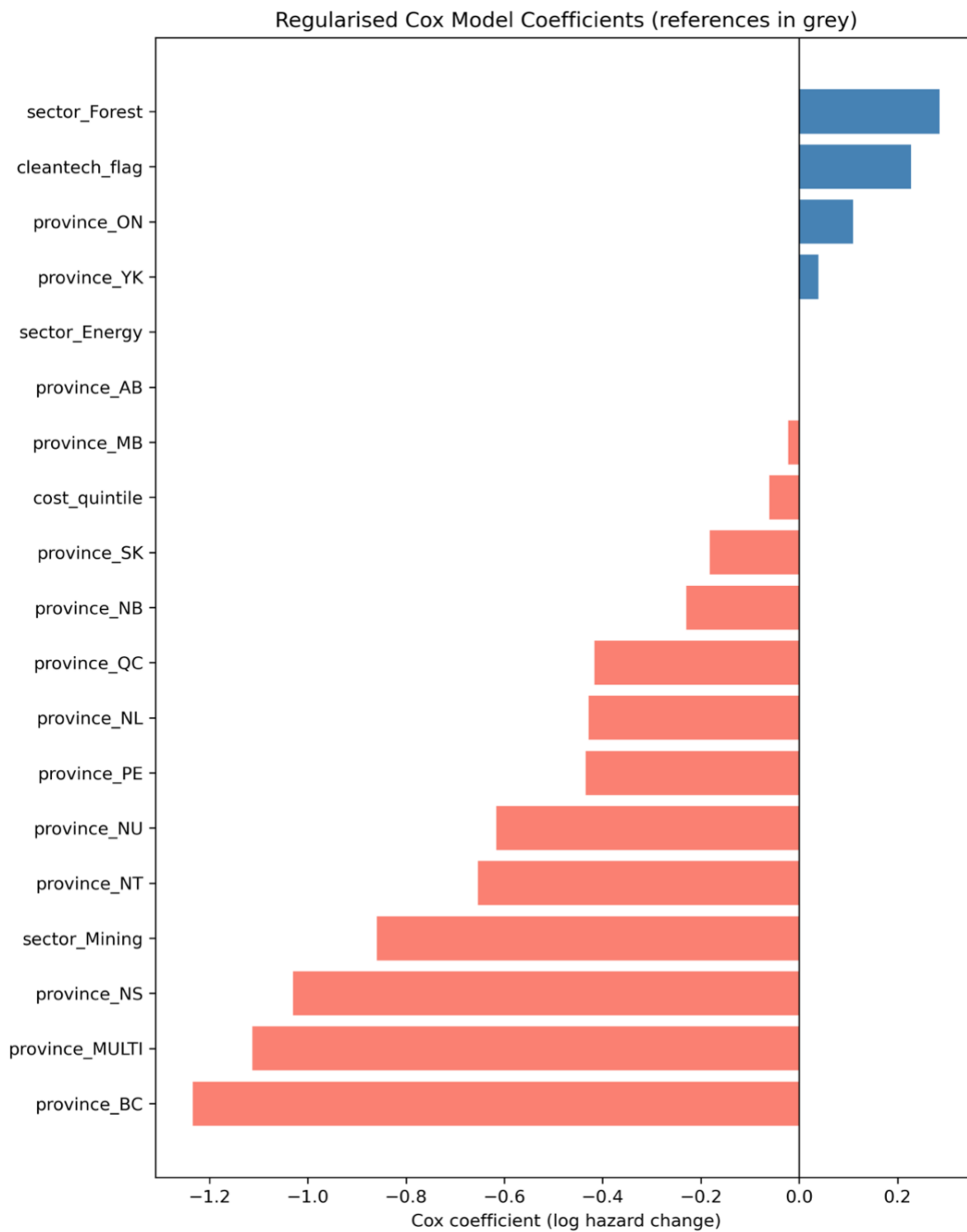


Figure 7: Regularised Cox Model Coefficients.

To convert these abstract hazards into planning guidance, the fitted model generates a survival curve for each project. The median completion date is the point where a project’s conditional survival probability intersects the 50 percent line. Subtracting years already spent in development converts this milestone into an intuitive “years remaining” estimate.

Table 1 returns to the two worked examples. For the Alberta coal mine (AB–Coal), the survival curve never crosses the 50 percent line, so the model returns a median of ∞ , signalling that at least half of comparable projects never reach resolution within the observation window. Its mean time-to-event is estimated at 5.2 years. Having already spent seven years in the MPI, the project is statistically overdue. By contrast, the Ontario precious metals mine (ON–PrecMetal) has a mean time-to-event of 5.0 years. At four years old, it retains about one year of average runway before a decisive outcome is expected. In both cases, hazard ratios (≈ 0.33 for AB–Coal and ≈ 0.39 for ON–PrecMetal) indicated slower progression than the Alberta–Energy baseline, but the survival analysis distinguished between a project that is effectively out of time and one that remains within its expected window.

label	province	sector	group	cost_quintile	cleantech_flag	beta	hazard_ratio
AB-Coal	AB	Mining	Coal	4	0	-1.10	0.33
ON- PrecMetal	ON	Mining	Precious Metal	3	0	-0.93	0.39

Table 2: Survival Log-Odds Scale for Two Worked Samples from Dataset.

Discussion

The results demonstrate that the Bayesian scorecard and Cox model not only meet functional accuracy thresholds but also preserved the interpretability required for project-level diligence and the outputs translate directly into planning guidance. They also emphasise the dominance of geography and sector as predictors of both probability and timing. Provincial context, particularly in combination with sector, exerted the strongest influence, while capital cost and cleantech status provided only modest additional lift. This pattern reinforces the importance of local context in project development, where permitting, infrastructure, and political climate strongly shape outcomes. Finally, the models also showed that many projects remain in the MPI well beyond the five-year median without resolution, highlighting the drag of “slow-moving” proposals that tie up policy attention and private capital without delivering results.

An important insight emerges when interpreting hazard ratios in a planning context. In project triage, an early exit, whether through timely construction or a swift cancellation, liberates capital, staff time, and institutional resources. Proposals with negative coefficients, which correspond to lower hazards, sit high on the survival curve and advance slowly. They drain resources even though their likelihood of success remains low. By contrast, projects with positive coefficients exhibit higher hazards, exit more quickly, and clear the pipeline sooner. This stands in sharp contrast to clinical research, where rapid attrition alarms physicians and long survival tails are the desired outcome. Context therefore dictates the value of hazard. In medicine, longevity is prized, whereas in infrastructure development, decisive outcomes, whether positive or negative, are more beneficial.

The models also illustrate the importance of interpretable methods in domains where accountability and auditability are essential. Likelihood ratios and hazard ratios are accessible to

policymakers and financiers without specialised training in machine learning, yet they meet functional accuracy thresholds comparable to more complex methods. This transparency facilitates recalibration and extension as new data become available, and it lowers the barrier for adoption across government and industry.

At the same time, several limitations warrant attention. The models assume coefficients are static over the seven-year window, an assumption that may not hold under shifting policy regimes or macroeconomic shocks. Sparse categories, particularly at the province–sector level, were stabilised with smoothing but may obscure genuine effects. Cost data, grouped into quintiles, may mask more granular scale dynamics. Finally, the proportional-hazards assumption, while generally satisfied, may understate the role of time-varying constraints such as regulatory approvals early in the process and financing bottlenecks late in development.

Despite these limitations, the paired models provide a transparent and auditable baseline for project surveillance and planning. By meeting functional accuracy thresholds, they demonstrate that relatively simple statistical methods, augmented with modern automation tools, can deliver insights comparable to those of far more complex approaches. For market participants and strategic planners, the key contribution lies in reframing Canada’s project pipeline as a dynamic risk surface: one that quantifies both the odds of success and the expected time to resolution, enabling more disciplined portfolio management and policy design.

Conclusion

This study set out to answer two practical questions posed by Canada’s Major Projects Inventory: “Will it build?” and “How much longer?” By pairing a Bayesian scorecard with a Cox proportional hazards model, we developed the first project-level toolkit that estimates both the

probability of construction and the expected time to resolution for major projects in Canada. The Bayesian scorecard achieved an AUC of 0.81 and the Cox survival model a C-index of 0.79, both clearing the mid-0.70s benchmark for functional accuracy.

The results demonstrate that geography and sector exert the strongest influence on project outcomes, while capital cost and cleantech status provide only modest additional lift. Used together, the models generate a unified risk surface that reveals both the likelihood and timing of construction for every active project in the MPI. The framework is lightweight, auditable, and quick to refresh, making it suitable for ongoing project surveillance and portfolio management.

The principal limitations lie in the independence assumptions of the likelihood ratios, the stationarity of coefficients, and the exclusion of macro-level shocks and micro-level considerations that may alter project viability. Future refinements could include rolling priors that update annually, finer geographic and stakeholder tags, integration of macroeconomic indicators, and a schedule for routine recalibration. An interactive dashboard would extend accessibility, allowing policymakers, financiers, and planners to visualise and stress-test project pipelines more directly.

By providing transparent, reproducible measures of both probability and timing, the models establish a shared evidence base for evaluating the fate of major projects in Canada. This approach closes a critical gap in market surveillance and strategic planning, shifting project analysis from judgment and anecdote toward auditable probabilities and timelines.